Neural Networks

Mathematics of AI - RPTU Kaiserslautern

WiSe 25/26

Topics of the lecture

- Neural Networks
  - MLP in `scikit-learn`
  - `pytorch`
  - NN training (variants and optimizers)
  - More complex networks (LeNet)

MLP in `sklearn`

As for previous methods, you can use a `sklearn` classifier:

```
MLPClassifier(
  hidden_layer_sizes=(10,10,), # or (10,)...
  activation='relu', # or `logistic`, `tanh`
  solver='sgd`, # or `adam` or `lbfgs`
  alpha=1e-3, 3 l2 regularization
  batch_size=32, # more on this later
  learning_rate_init=1e-3, # learning learning_rate_init
  ...
)
```

MLP in `pytorch`

In `sklearn`, you don't have too much possibility to define custom networks (e.g., to operate on images).

Those networks should be manually implemented, as they can be very domain-specific. `pytorch` is a library for simplifying this process.

In notebook `1_MLP_torch.ipynb` you find a walkthrough on how to implement a MLP using `torch`.

NN training

Training of a neural network :

- evaluate model on a set (X, y) to obtain $\hat{y}$
- calculate loss ($\mathcal{L}(\hat{y}, y)$)
- compute gradient of loss w.r.t parameters $\nabla_\theta \mathcal{L}$
- improve the parameters

$$\theta^{(i+1)} = \theta^{(i)} - \alpha \nabla_\theta \mathcal{L}$$

Variants

In notebook `2_stochastic_gradient_descent.ipynb`, we will explore variants of SGD and optimization algorithms.

Variants:

- SGD : Let the set be one sample at a time
- Mini-batch : Let the set be “some” samples (not all)
- GD : Calculate gradient on the entire data set

Optimizers

Different optimizers can be used to opdate the parameters.

For example, momentum can be added:

$$v^{(t+1)} = \mu v^{(t)} - \alpha \nabla_\theta \mathcal{L}\left(\theta^{(t)}\right)$$

$$\theta^{(i+1)} = \theta^{(i)} - v^{(t+1)}$$

and more (next semester's topic).

See <u>this visualization</u> and <u>and this comparison</u>.

LeNet in pytorch

Notebook `3_lenet_torch.ipynb` contains an implementation of LeNet, to be trained on the MNIST dataset.

This notebook is more resource-intensive, and downloads a large dataset from the internet. **Make sure to delete the dataset if you don't need it anymore**.